Ultra-Low-Resolution Thermal Sensing Can Estimate Walkable Directions Without Recalibration or Manual Labels

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Motivation: Wearable navigation still depends on detailed vision, calibration, and manual labels

- Wearable aids typically rely on high-res cameras and depth sensors, raising privacy concerns.
- Building such wearable systems requires costly datasets where humans manually label the walkable direction in each image.
- Most systems need geometric calibration and a fixed camera layout, which breaks if a sensor shifts.
- **Can low-resolution thermal sensors determine whether the path to the left, center, or right is clear, without manually labeling thermal images or recalibrating the system when a sensor moves?**

Methods: RGB auto-labels supervise training, so deployment needs only the thermal cameras

- A wearable rig with three low-res thermal cameras captures multi-view data while walking.
- A high-res RGB camera, used only during data collection, auto-generates clear/person/object labels.
- At inference, the model needs only the thermal tiles - no RGB camera, no fixed layout. Cameras are aligned once, at assembly, so RGB labels match the right thermal camera during training.
- Collected around the CMU-Q campus, indoors and outdoors; 27,852 RGB-thermal frames, with 15,500 used for training; vs. early/late fusion.

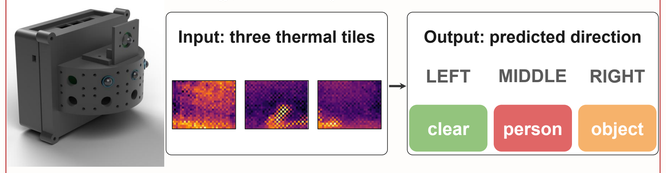


Fig. 1: *The wearable rig, its three thermal cameras, and the direction it predicts from them.*

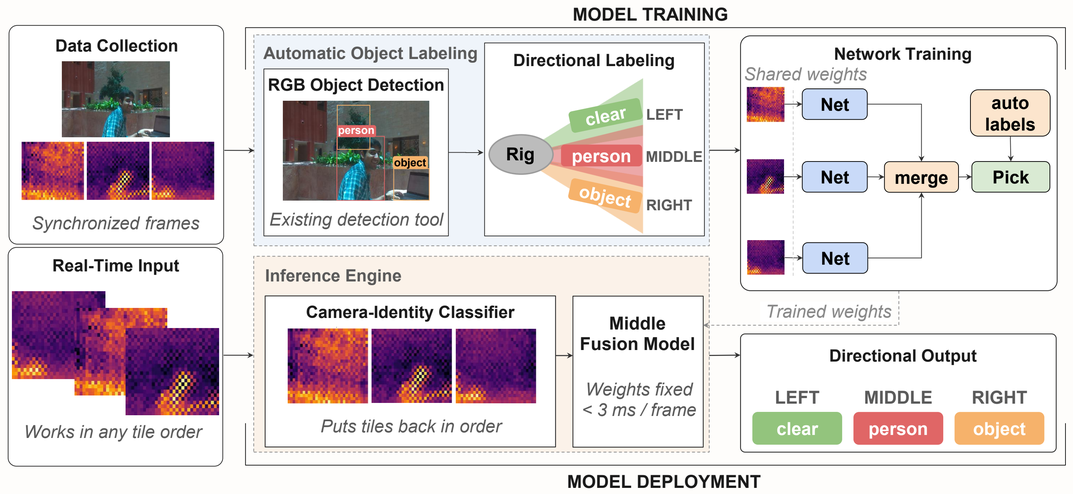


Fig. 2: *End-to-end pipeline: RGB/YOLOv8 auto-labels supervise training; at deployment, re-sorted thermal tiles feed the fusion model*

Finding 1: Middle fusion beats early and late fusion

- Middle fusion leads for every class: overall per-class F1 score (a standard measure balancing correct detections against false alarms) of 0.81 (clear), 0.83 (person), 0.80 (other object), vs. 0.73 - 0.78 for the other two methods.
- Combining features at every layer of the network preserves spatial detail that the other methods lose.
- Running the extra network costs more compute, yet every model still runs in under 3 ms per frame on CPU.

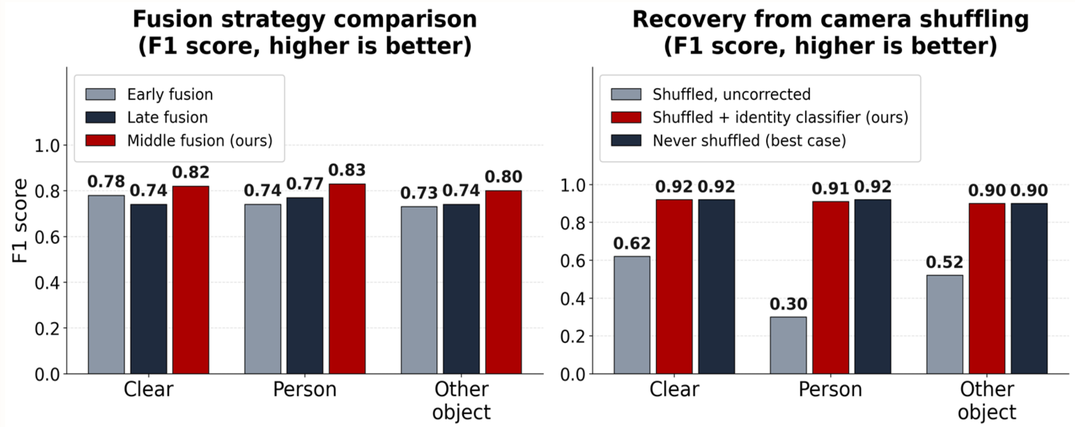


Fig. 3: *Left: F1 by class for each fusion strategy. Right: F1 after camera shuffling, with and without the identity-classifier correction, compared to a never-shuffled system.*

Finding 2: The system stays accurate even as the cameras move

- Shuffling camera slots collapses uncorrected Middle Fusion: F1 drops to 0.62 (clear), 0.30 (person), and 0.52 (other object).
- A lightweight camera-identity classifier re-sorts the tiles with 98.7% accuracy, restoring F1 to 0.90 - 0.92 - almost matching an unshuffled system, at only +0.21 ms.
- What breaks under shuffle is which camera feeds which slot, not the thermal images themselves.

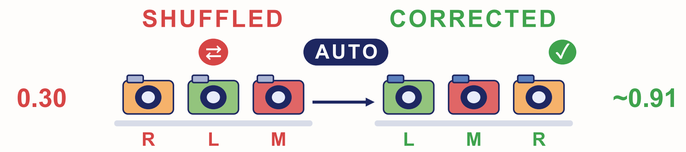


Fig. 4: *When a camera moves, the classifier re-identifies it and restores accuracy automatically.*

Implications: Calibration-free, ultra-low-res sensing broadens access

- Each thermal camera captures only a 32x24-pixel array - too coarse to identify anyone, yet enough for navigation.
- No manual labels and no per-deployment calibration lower the barrier to entry: the full rig costs ≈ $471.
- **Main contribution:** useful navigation guidance from cheap, privacy-preserving sensors - with no manual labeling and no recalibration when the sensors move - plus a public 27,852-frame RGB - thermal dataset.

References

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